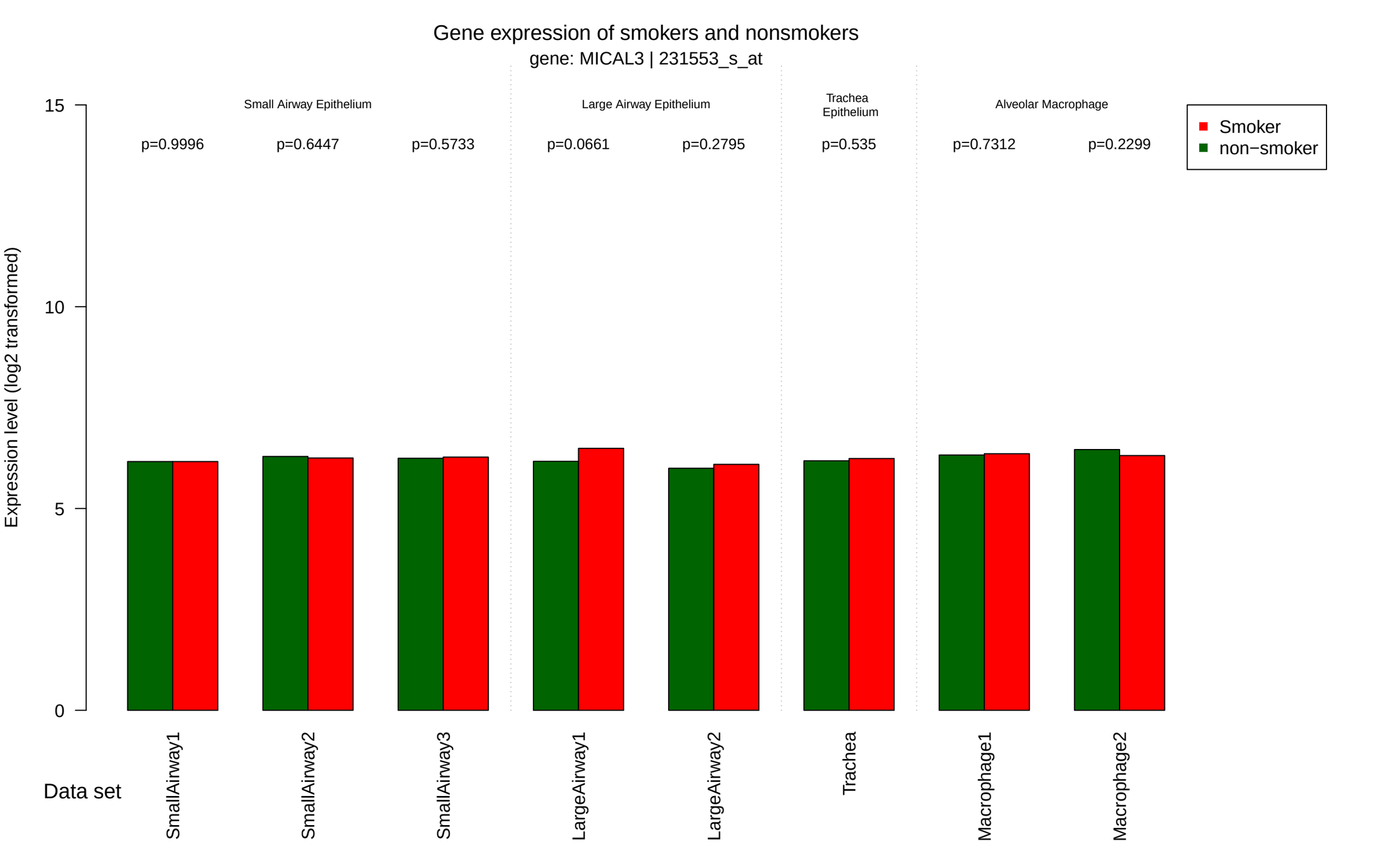
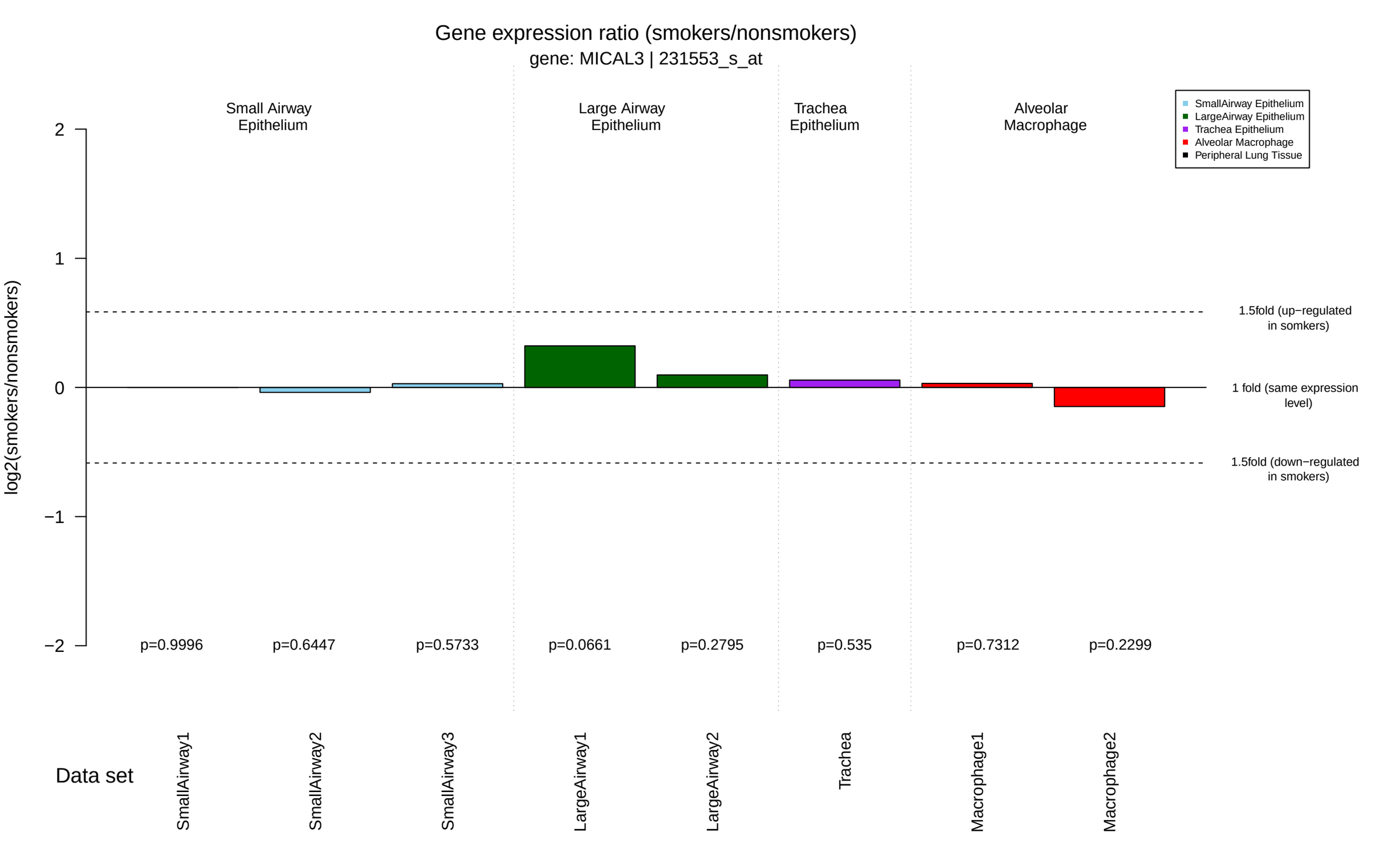
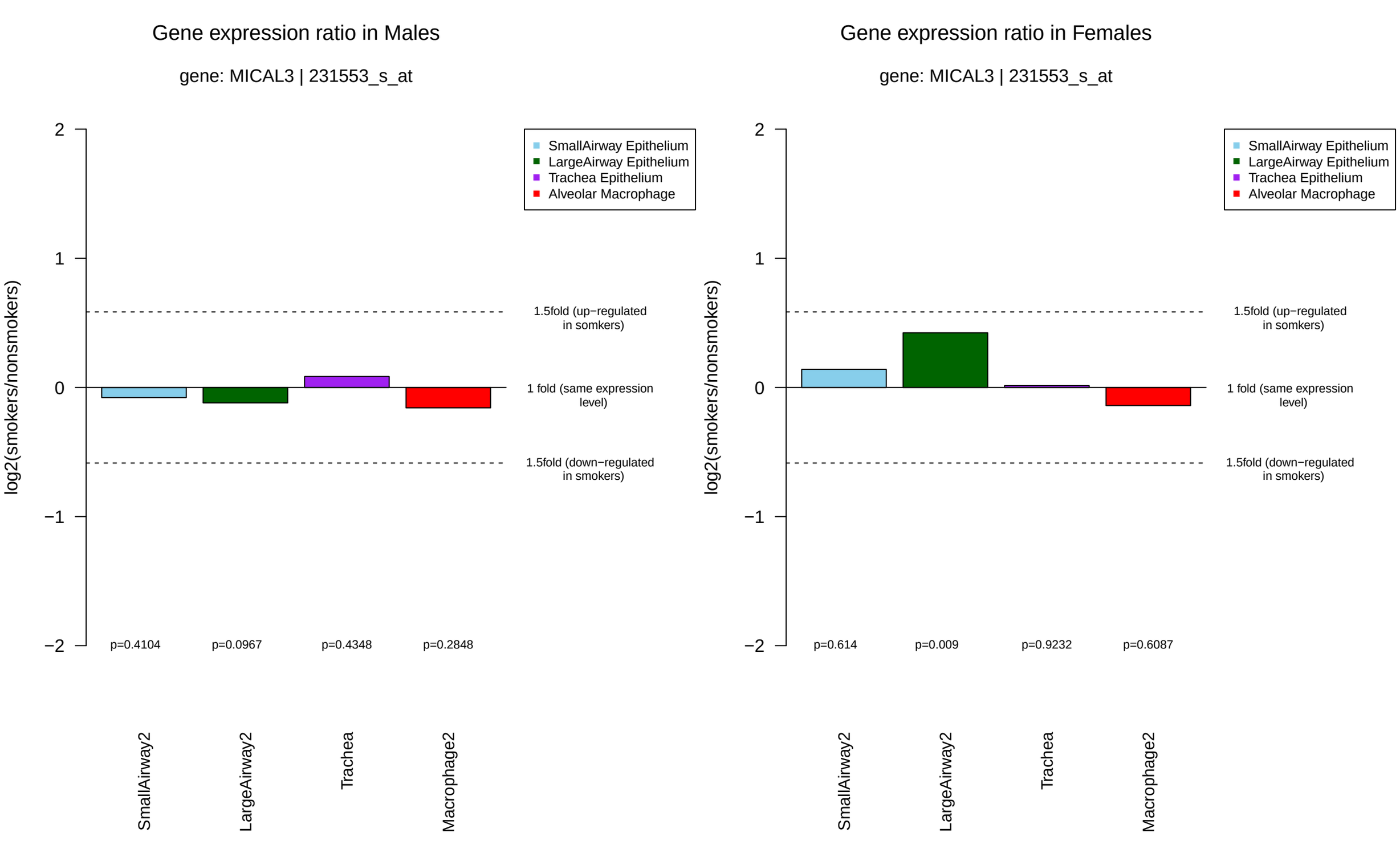
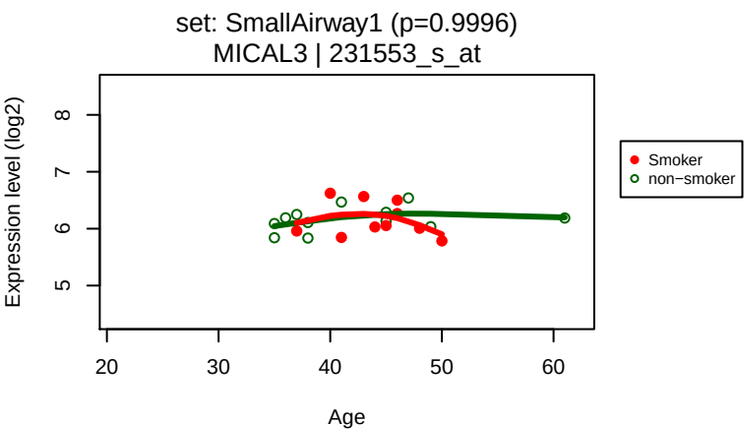
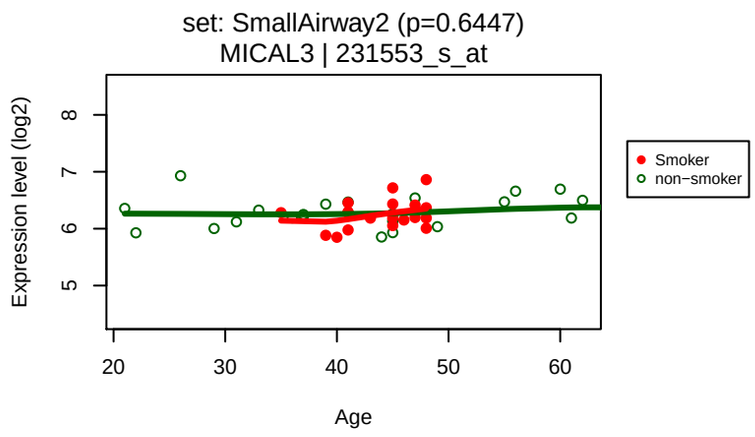
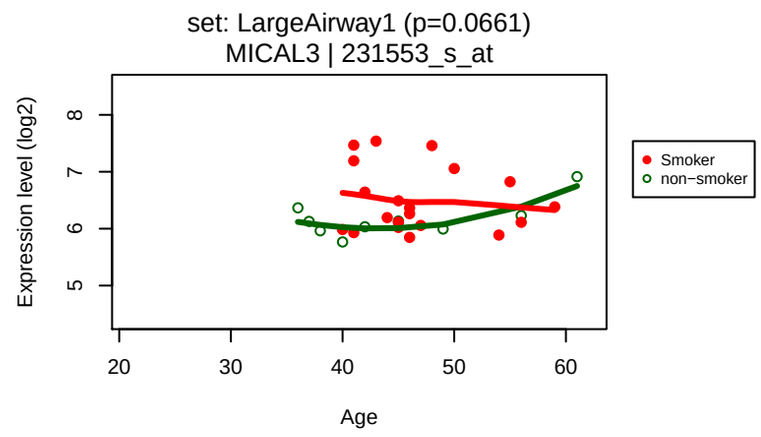

Gene expression of smokers and nonsmokers
gene: MICAL3 | 231553_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9996
p=0.6447
p=0.5733
p=0.0661
p=0.2795
p=0.535
p=0.7312
p=0.2299
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

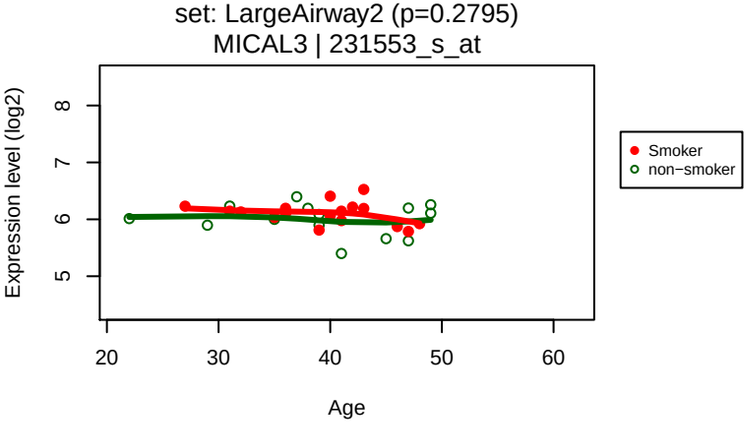

Gene expression ratio (smokers/nonsmokers)
gene: MICAL3 | 231553_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9996
p=0.6447
p=0.5733
p=0.0661
p=0.2795
p=0.535
p=0.7312
p=0.2299
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

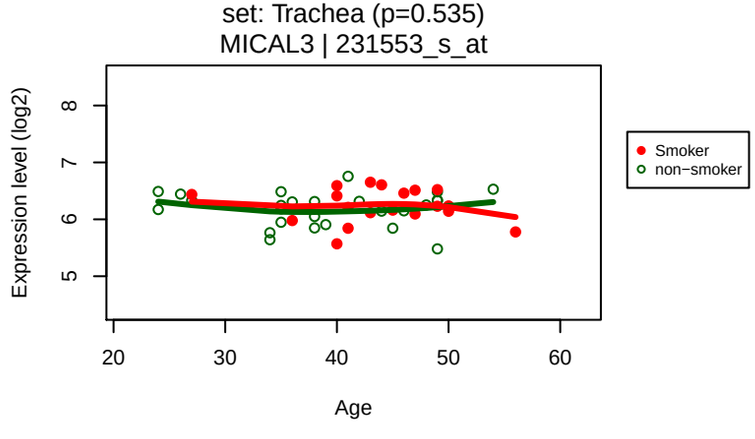

Gene expression ratio in Males
gene: MICAL3 | 231553_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4104
p=0.0967
p=0.4348
p=0.2848
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MICAL3 | 231553_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.614
p=0.009
p=0.9232
p=0.6087
SmallAirway2
LargeAirway2
Trachea
Macrophage2

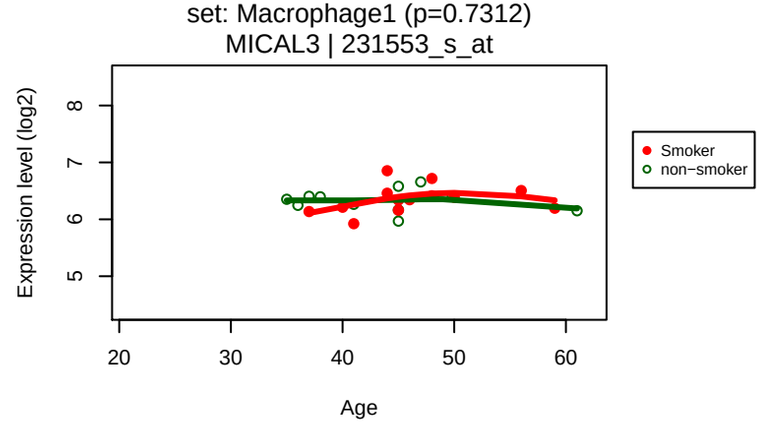

set: SmallAirway1 (p=0.9996)
MICAL3 | 231553_s_at
Expression level (log2)
Age
Smoker
non−smoker

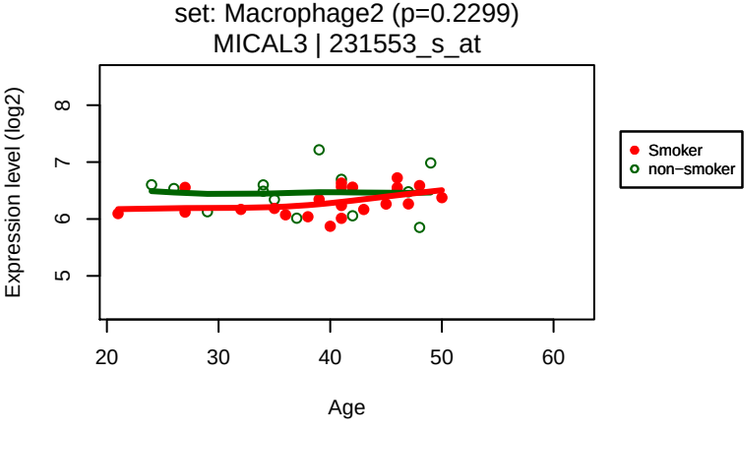

set: SmallAirway2 (p=0.6447)
MICAL3 | 231553_s_at
Expression level (log2)
Age
Smoker
non−smoker

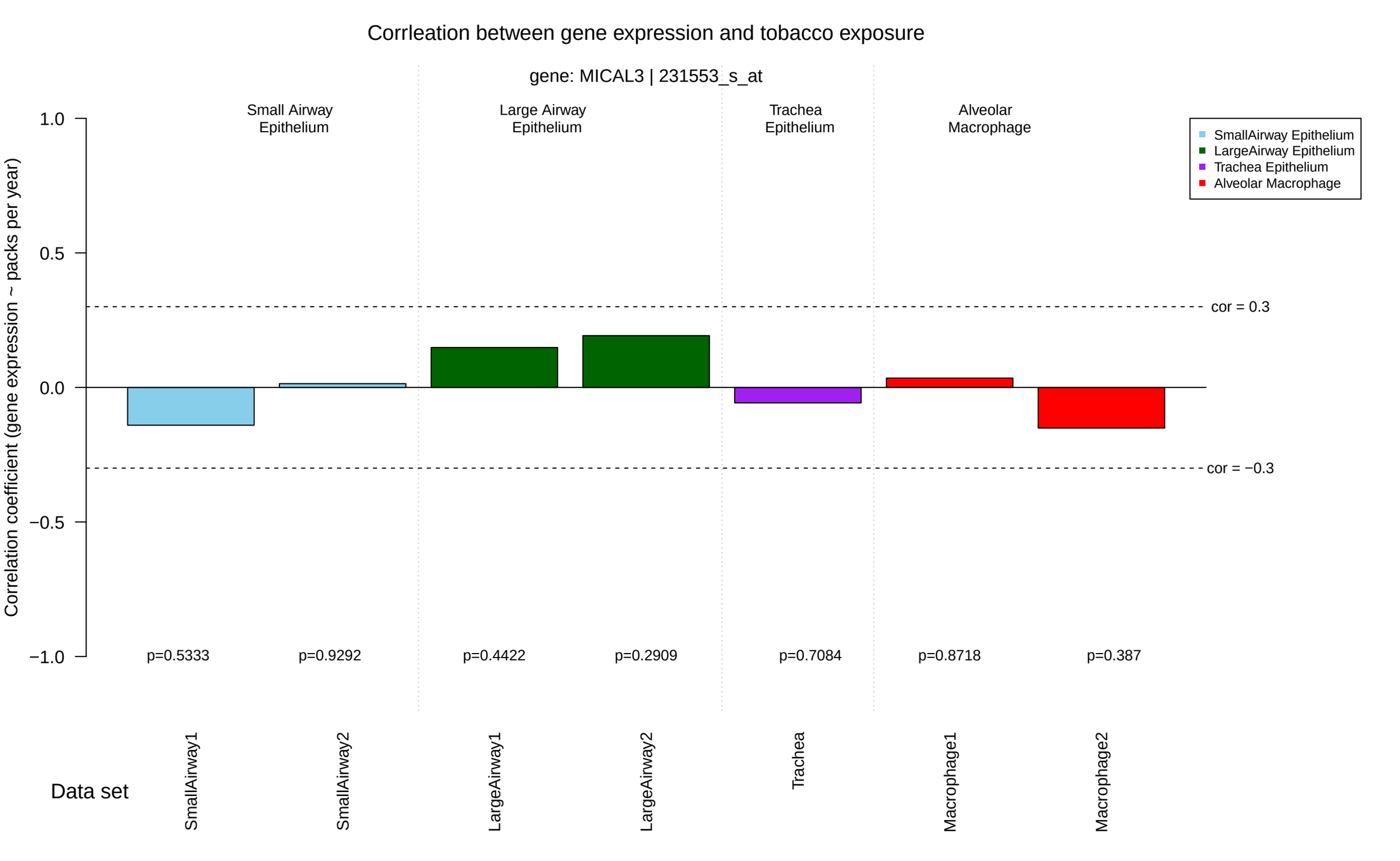

set: LargeAirway1 (p=0.0661)
MICAL3 | 231553_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2795)
MICAL3 | 231553_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.535)
MICAL3 | 231553_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7312)
MICAL3 | 231553_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2299)
MICAL3 | 231553_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MICAL3 | 231553_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5333
p=0.9292
p=0.4422
p=0.2909
p=0.7084
p=0.8718
p=0.387
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2